“Gas Leakage Detection System”
A project Submitted to the
School of Computer Science of

Yashwantrao Chavan Maharashtra Open University, Nashik,
in Partial Fulfillment of
the Requirements for the Degree of Bachelor of Science (BIS)/ B.C.A

Submitted By
Ansari Mohammad Arif Azam Ali (2022017000797222)

Academic Year (2024-2025)
K.M.E. SOCIETY’s G.M MOMIN WOMEN’s COLLEGE
RAIS HIGH SCHOOL CAMPUS OF KOKAN MUSLIM EDUCATION
SOCIETY (35226)

A

PROJECT REPORT ON “GAS LEAKAGE DETECTION SYSTEM”

SUBMITTED TO

YASHWANTRAO CHAVAN MAHARASHTRA OPEN UNIVERSITY

IN PARTIAL FULFILLMENT FOR THIRD YEAR OF BACHELOR OF COMPUTER APPLICATIONS (BCA)

SUBMITTED BY

Ansari Mohammad Arif Azam Ali (2022017000797222)

UNDER THE GUIDANCE OF

Prof. Agha Safa

THROUGH

K.M.E SOCIETY’S G.M MOMIN WOMEN’S COLLEGE
RAIS HIGH SCHOOL CAMPUS,
OLD THANE ROAD, BHIWANDI,
THANE 421302

2024-2025

YCMOU STUDY CENTRE 35226
K.M.E. Society's G.M. Momin Women's College,
RAIS HIGH SCHOOL CAMPUS, OLD THANE ROAD, Bhiwandi, THANE-421302
raishsjc@gmail.com/amfccenter@gmail.com ,9607175524/9422570462

Date:

Certificate of Evaluation

This is to certify that the undersigned have assessed and evaluated the Project Work titled **"Gas Leakage Detection System"**

1) Ansari Mohammad Arif Azam Ali (2022017000797222)

The Project Report has been (accepted/rejected) for the partial fulfillment of BCA Programme

Signature of the Examiner

Name of the Examiner

Stamp of the Study Centre

YCMOU STUDY CENTRE 35226
K.M.E. Society's G.M. Momin Women's College,
RAIS HIGH SCHOOL CAMPUS, OLD THANE ROAD, Bhiwandi, THANE-421302
raishsjc@gmail.com/amfccenter@gmail.com ,9607175524/9422570462

Date:

Certificate of Completion

This is to certify that the team of the following students of BCA have completed the project work titled " **Gas Leakage Detection System "** under my guidance and supervision. The project report has been written according to the guidelines given by the Yashwantrao Chavan Maharashtra Open University

1) Ansari Mohammad Arif Azam Ali (2022017000797222)

Signature of the study Center Coordinator

Signature of the Guide

Name of the study Center Coordinator

Name of the Guide

Stamp of the study Centre

CERTIFICATE OF ORIGINALITY

Date:

This is to certify that the project entitled "**Gas Leakage Detection System**" Submitted to the School Of Computer Science in partial fulfillment of the requirement for the award of The degree of BACHELOR OF COMPUTER APPLICATIONS (Yashwantrao Chavan Maharashtra Open University), is an original work carried out by

Name: Mr. Ansari Mohammad Arif Azam Ali
PRN No: 2022017000797222
Exam Seat No:

Under my guidance. The matter embodied in this project is a genuine work done by the student and has not been submitted whether to this Organization or to any other University/Organization for the fulfillment of the requirement of any course of study.

Signature of the Internal Guide
Prof. Agha Safa

Signature of the Student
Mr. Ansari Mohammad Arif Azam Ali

CERTIFICATE

This is to certify that the project report entitled "**Gas Leakage Detection System**" Submitted to the School of Computer Science in partial fulfillment of the requirement for the award of the Degree of BACHELOR OF COMPUTER APPLICATIONS (Yashwantrao Chavan Maharashtra Open University) is an original work carried out by

Name: Mr. Ansari Mohammad Arif Azam Ali

PRN No: 2022017000797222

Exam Seat No:

The matter embodied in this project is a genuine work done by the student and has been Certified by the following external examiners deputed by Yashwantrao Chavan Maharashtra Open University.

External Examiner

Role and Responsibility

The main role of the Gas Leakage Detection System is for the sake of loves safety and fulfilment of social duties, and keeping in focus the life- threatening instances of blasts and injuries due to leakage of gas in industries, vehicles and houses. A gas leakage system is designed whereby application of embedded system and involvement of Internet of Things (IoT). The main objective of the work is designing micro controller based toxic gas detecting and alerting system.

The Gas Leakage Detection System and acquire project that will perceive gas outflow like Methane leak, Butane leak and LPG leak, Methane overflow or any such petroleum centered on gaseous substances that can be discovered using MQ135 device to layout and set up an SMS alert missive to restrict mobile number enter inside the Arduino Program. to layout and acquire a project that will fabricate a sound alarm during gas outflow is regulated.

ACKNOWLEDGEMENT

I would like to express my heartfelt gratitude to all those who have supported and guided me throughout the development of my project titled “Gas Leakage Detection System”.

First and foremost, I extend my sincere thanks to **Mrs. Agha Safa**, my project guide, for her invaluable guidance, encouragement, and insightful feedback. Her support was instrumental in navigating the challenges encountered during this project.

I have been greatly benefitted from their regular critical reviews and inspiration throughout my work. I am grateful to them for their guidance, encouragement, understanding and insightful support in the development process. I am deeply appreciative of the **BCA** staff of our college for their constant encouragement, assistance, and for providing the necessary resources that facilitated the successful completion of this project.

My sincere thanks to Yashwantrao Chavan Maharashtra Open University (YCMOU) for offering a robust academic platform that enabled me to undertake this project. The university’s comprehensive curriculum and supportive environment have significantly contributed to my technical and professional growth.

I am also thankful to entire staff of **BCA** for their constant Encouragement, suggestions and moral support throughout the duration of my project. Last but not the least I would like to mention here that I am greatly indebted to each and everybody my friends and who has been associated with my project at any stage but whose name does not find a place in this acknowledgement.

TABLE OF CONTENTS

Sr.no.	Topic	Page no	Remark
1	**Introduction** 1.1 Background 1.2 Objective 1.3 Purpose, scope, applicability 1.4 Achievements 1.5 Organizing of report	1-6	
2	**Survey of the technology** 2.1 Existing system 2.2 Proposed system	7-10	
3	**Requirements and Analysis** 3.1 Problem definition 3.2 Requirements specification 3.3 Planning and scheduling 3.4 System analysis 3.5 Feasibility study 3.6 Prototyping model	11-22	
4	**System design** 4.1 Data design 4.2 Algorithm 4.3 Testing period 4.4 Gantt chart 4.5 summary	23-33	

5	**IMPLEMENTATION AND TESTING** 5.1 Implementation Approaches 5.2 Coding Details 5.3 Code Efficiency 5.4 TestingApproach 5.5 Modifications and Improvements 5.6 Test Cases	34-50	
6	**RESULTS AND DISCUSSION**	51-55	
7	**CONCLUSION AND FUTURE WORK**	56-59	

List of Figures

SR no.	Figures	Page no.
1	Arduino Uno	13
2	Arduino Uno Pinout	15
3	MQ 135 Sensor	16
4	Arduino IDE	17
5	Prototype	22
6	Data flow diagram	25
7	Block diagram	26
8	Circuit Diagram	27
9	Algorithm diagram	28
10	Gantt Chart diagram	30
11	Sequence diagram	31
12	Use case diagram	32
13	Life cycle diagram	35
14	MVC Architecture	40
15	Testing Strategy diagram	44
16	Test results	49
17	Screenshot of Before applying the gas leakage	52
18	Screenshot of After applying the gas leakage	53
19	The graph of Gas Alarm/detection response	55

1. INTRODUCTION

INTRODUCTION

With the development of world, the technology is enhanced day by day with the realistic projects and efficient work. In this project we are using MQ semiconductor sensor to detect combustible gas. This sensor has lowered conductivity in fresh air. Conductivity of this sensor depends upon concentration of the gas so it may detect not only combustible gas but also smoke, butane, isobutene, liquor. If sensor senses the outflow from storage, then the output goes low. This low signal is monitored by the microcontroller and it will establish the gas outflow. Currently the microcontroller turns on LED and Buzzer, it conjointly activates fan for throwing gas out and continue send messages as "ALERT!! GAS LEAKAGE" to your mobile number subsequentially the stepper motor turns off the switch of the cylinder.

More and more people are getting injured, burnt and a large portion of them Eventually dying due to cylinder blasts that have been taking place in the country for the last couple of years. At least 50 were injured and 10 died in last one-year time in Bangladesh. In the last January alone, there were 4 huge cylinder blast incidents that took place in the city.

According to the reported According to the reported incidents, cylinders are blowing out on a regular basis throughout the country. The frequency of the blast is almost one at least in a month. From the last June, there were at least 5-6 LPG and 2 CNG cylinders explosion incidents that are reported at various dailies. Those explosions cost at least 7 human lives and left 5 burnt and injured severely while the financial loss was above 30 to 35 million rupees. Experts have urged on setting up policy by the government to maintain the safety standards in this sector. They are also encouraging the mass people to be cautious while buying and using the cylinders and to conduct regular health check-ups of the cylinders from a certified authority

Abstract

Safety plays a major role in today's world and it is necessary that good safety systems are to be implemented in places of education and work. This work modifies the existing safety model installed in industries and this system also be used in homes and business premises. One of the preventive measures to avoid the danger associated with gas leakage is to install a gas leakage detector at vulnerable locations.

A gas detector is a device that detects the presence of gases in an area, often as part of a safety system. Gas Detector where it can sound an alarm to operators in the area where the leak is occurring, giving them the opportunity to fix or leave. This type of device is important because there are many gases that can be harmful to organic life, such as humans or animals. Containment into any area where the gas should not be present must be avoided. Because a small leak may gradually build up an explosive concentration of gas, leaks are very dangerous.

Nowadays, existing gas detector is less effective in usage because the user can only detect the gas leakage when they test by using gas detector. It is dangerous since gas leakage must be identified from early of the leak. That is Why the Gas Leakage Detector with Notifier System was invented to avoid the fire or explosion occur in the houses or premises. This kind of gas detector will detect the gas continuously as long as there is power supply.

This project used Microcontroller Arduino UNO at the processor where it processes the input from the sensor and to GSM module to communicate with the user by sending an alert through SMS. The buzzer will ring until its dangerous concentration of gas is achieved. The benefit of these projects is to prevent the earlier stage of fire because of unattended cooking without a human supervision, could prevent the explosion because of gas leakage.

Background

LPG is the terminology for Liquefied Petroleum Gas. The gas is made up of hydrocarbon gases compromising of Propane and Butane. LPG is prepared by refining petroleum or “wet” natural gas streams as they emerge from the ground.

As it is a gas, it does not pose ground or water pollution hazards, but it can cause air pollution. As its boiling point is below room temperature, LPG will evaporate quickly at normal temperatures and pressures and is usually supplied in pressurized steel vessels. LPG is also colorless and odorless. However, for safety reasons, an odorant is added to LPG for easy detection of any gas leakage.

LPG is heavier than air, unlike natural gas, and thus will flow along floors and tend to settle in low spots, such as basements. There are two main dangers from this. The first is a possible explosion if the mixture of LPG and air is is within the explosive limits and there is an ignition source. The second is suffocation due to LPG displacing air, causing a decrease in oxygen concentration.

Lately, gas leakage problem is highly become current issues that be the main causes of fire burning. This project is produced to detect any leakage of cooking gas and it will alert the user about leakage.

Gas detection of a hydrocarbon gas via infrared (IR) absorption requires the absorption of optical energy (IR) by the gas at the wavelength of interest. Different gaseous have different absorption spectrum. In this project, attention will be given to Liquefied Petroleum Gas (LPG) which is an energy source primarily composed of propane and butane. Due to the unique absorption properties of gas to infrared radiation, leakage can be detected by measuring and comparing the IR intensity at both source and detector. In order to enhance the effectiveness of the gas leakage detector, a circuit consisting of an alarm system will be implemented to the prototype to warn users when gas leakage occurs

Objective

- The primary objective of the present project is to provide a means for safely detecting any malfunction of a pressurized gas system in order to prevent accumulation of combustible gases is prevented.
- Another objective of the project is to provide safety means for detecting the leakage of gas into the area of an appliance when the appliance is in a shutdown condition and not in operation.
- Yet another objective of the gas leakage detection is to provide a leakage and monitoring system which is economical to manufacture and which may be readily installed in conventional trailers, boats, or like which are normally dependent upon a stored supply of pressurized gas
- Other than that, student must also add some of the safety actions which are the implementation of alarm system to alert users of the leakage occurrence. The alarm will trigger as soon as the infrared radiation has shown a positive detection when the leakage of LPG had occurred. Due to the trigger of the alarm, the air-valve is also activated to close the pipeline which in this prototype case, the plastic tube acted like one. Once the air valve has fully-closed, an alarm will be disabled.
- To implement a gas tank leakage detector which complete with alert alarm and screen monitoring system that can show the specific place of gas leakage and and at the same time the person on duty can fix the leak as soon as possible before it getting worst.

Purpose, scope and Applicability

Purpose

The main purpose of the project is intended to use in applications in industries or plants that depend on gas and liquified petroleum gas and natural gas in their operation. The is also can be used in household safety where appliances and heaters that use gas may be source of risk.

The hazardous gas that leaks from a lab or research center or power plant such as methane, butane can be very dangerous for humans and can blast the whole organization or an area. So, for safety measures if a Gas Leakage Detector can used to prevent disasters or provide a safe working environment for humans.

SCOPE

The scope of this project is to design a system that can detect the presence of natural gases. It can reduce the fire accident and explosion because of the gas leakage. Beside that it also can detect the early stage of gas leakage. This project will only focus for household. It will detect LPG gas.

In this project it uses a sensor that can detect LPG gas, propane, butane and smoke. This project used Microcontroller Arduino UNO at the processor where it processes the input from the sensor and to GSM module to communicate with the user by sending an alert through SMS.

Each thing in a world has a limit same goes to this project. This project limitation includes:

- The sensor-the gas and smoke sensor it will only detect LPG Gas (Butane/Propane) for gas
- The GSM module – the range it will cover only in the same city of the module SIM card. The roaming coverage will not include.
- The GSM module its interface through Short Message System. The module will not able to call the user.

Applicability

Gas leakage problem is highly become current issues that be the main causes of fire burning. This project is produced to detect any leakage of flammable gases and it will alert the user about the user about leakage. User will get the alert in SMS. By using this project, it will reduce the accident of fire and explosion. It also helps the early detection of gas before the concentration of gas reaches the dangerous level.

Achievements

Once the project is completed the gas detection and monitoring system can be used to monitor and test the leakage of pressurized gas and can be used in industries to prevent the disasters because of the leakage of hazardous gaseous substances. In household uses it can detect leakage and also it is connection with the network and sends message alert to the registered mobile number so that the user gets the alert about the leakage at home.

After the completion of the project one can get the complete understanding of software project management and the project development lifecycle under which the project is developed. The project also trains the developer in Internet of Things (IoT) development. It gives developer the complete understanding of microcontrollers such as Arduino and Raspberry pi. And different sensors and actuators used in project development.

Organization of Report

Gas Leakage Detector with Notifier System is device that that detects the Presence of gases in an area, often as part of safety system. It detects gas leakage continuously. When leakage occur, it will alert the user about leakage in SMS. The buzzer will ring to give alert to the people, who are there in the house or premise.

One of the preventive measures to avoid the danger associated with gas leakage is to install a gas leakage detector at vulnerable locations. Gas leakage problem is highly become current issues that be the main causes of fire burning. This project is produced to detect any leakage of cooking gas and it will alert the user about leakage. User will get the alert in SMS. By using this project, it will reduce the accident of fire and explosion. It also helps the early detection of gas before the concentration of gas reaches the dangerous level.

2. SURVEY OF TECHNOLOGIES

SURVEY OF TECHNOLOGIES

This chapter will provide a review about the previous research and existing project that have been made by using reference sources and guidelines as journals, internet, article writing, blog and scientific studies to get an idea about the project design, conception and any information that related to improve the project. With a differences concept and design, there are other creation and innovation of projects done by the other people. The research that is related to this project also covered in this chapter.

Gas Leakage Detector with Notifier System is the innovation from existing Gas Detector. This project is mainly for an individual who wants to keep their home safe from the risk of burning. The benefit of this project is to detect any leakage of cooking gas and it will alert the user about leakage. Microcontroller Arduino UNO at the processor where it processes the input from the sensor and to GSM module to communicate with the user by sending an alert through SMS. This kind of gas detector is more efficient as the user will get the information faster and the detection of gas is continuous.

Gas detectors measure and indicate the concentration of certain gases in an air via different technologies. Typically employed to prevent toxic exposure and fire, gas detectors are often battery-operated devices used for safety purposes. They are manufactured as portable or stationary (fixed) units and work by signifying high levels of gases through a series of audible or visible indicators, such as alarms, lights or a combination of signals.

While many of the older, standard gas detector units were originally fabricated to detect one gas, modern multifunctional or multi-gas devices are capable of detecting several gases at once. Some detectors may be utilized as individual units to monitor small workspace areas, or units can be combined or linked together to create a protection system.

As detectors measure a specified gas concentration, the sensor response serves as the reference point or scale. When the sensors response surpasses a certain pre-set level, an alarm will activate to warn the user. There are various types of detectors available and the majority serves the same function to monitor and warn of a dangerous gas level

Existing System

Portable Multi Gas Detector (XA-4400II)

It is a multi-gas detector which can monitor combustible or flammable gases including LPG and methane, hydrogen sulphide/sulfide, carbon monoxide, and oxygen. This comes with an attachable pump.

Super Sensitive Portable Gas Leak Detector (XP-702III Series)

These portable gas detectors are highly sensitive, making them an effective tool when it comes to locating trace gas leaks in a rather shorter time and offer a remarkably more sensitive solution to any leak detection liquid. This is equipped with a built-in gas sampling pump, and depending on the specific model you choose, these detectors are available in both dual and single gas applications.

Issues with existing gas detector

Faulty gas detection instruments can be fatal. There are many factors in the workplace that can cause a gas detector to fail. Here are the issues:

1. **Environmental**

   Dirt, dust and water impact. These physical affects can block gases and vapors' from entering the sensor chamber preventing detection of the gases.

2. **Physical Affects**

   Dropping and other abuse can damage the instrument from working properly or at the least change the ability of the detector from measuring accurately.

3. **Gas Exposure**

   High gas exposure will change the calibration curve of the sensors causing false or inaccurate readings. Extremely high concentrations can kill the sensor's ability to measure gas.

4. Temperature Affect

Storing instruments in environment which is either too cold or too hot can affect the ability of the sensors to measure accurately.

5. Moisture

Moisture condensing on or in sensor: this can happen to oxygen sensors when moisture condenses in the capillary tube in the sensor. It will cause the sensor to fail.

6. Calibration Drift

All sensors from all manufacturers drift over time. Calibration brings the sensor back into equilibrium and provides accurate readings.

Proposed System

The project entitled "Safety Helmet for Gas Leakage using Arduino with Sound Alarm", will be a great help in terms of preventing any danger caused by gas leakage. The purpose of this project is to detect the presence of LPG leakage as a part of a safety system. Apart from sound alarm, a Message alert will inform the authorized person and the solenoid valve will be triggered to shut down the gas supply to prevent any harmful effects due to gas leakage.

Descriptively, we use a gas sensor to monitor the LPG if the gas leak reaches beyond the normal level. This proposed project will trigger the sound alarm. In addition, the authorized person will be informed about the leakage via SMS alert and the gas supply will be automatically shut down. The people can be saved from a potential explosion caused by gas leakage.

Expectations is when the system should be capable of giving a warning of both the presence and the general location of an accident accumulation of flammable gases, in order to initiate one or more of the following actions, either automatically or under manual control. This system is complete with IOT system which is provide better performance which can monitor and fix easily. This system also can be monitored remotely.

3. REQUIREMENT AND ANALYSIS

REQUIREMENT AND ANALYSIS

Problem Definition

In this section we define the problems. The problem is that collected source data is properly work or not. Defining the problems that are comes under the project. There are no standard recommendations or specific guidelines mentioned regarding the hardware requirements for creation of Gas Leakage Detection System. The minimum requirement is a Arduino uno board, GSM module and MQ sensor

Requirements Specification

In this section we include the requirements of the project. The components required to develop the project named “Safety Helmet for Gas Leakage” are as follows:

Hardware Requirements:

- Arduino Uno R3.
- MQ135 gas sensor.
- Buzzer.
- LED light.
- Battery.
- Jumper wires.
- GSM Module.

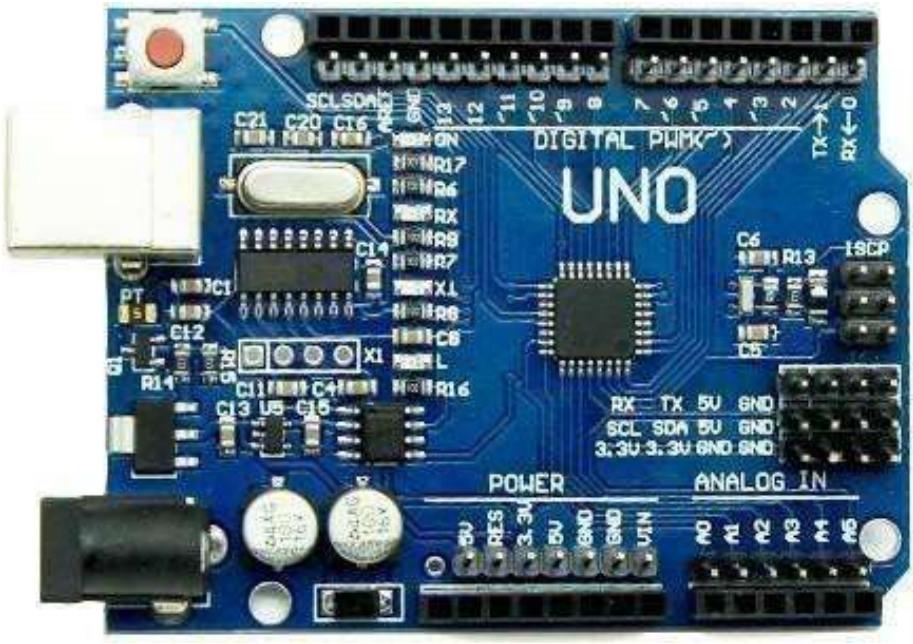
Software Requirements:

- Arduino IDE.
- C++ Language.

Arduino UNO

Microcontroller Arduino UNO is the central unit of the system is Arduino board, where all the components are interfaced externally on the board and programmed as per their functionality to work in synchronization. It is an electronic prototyping platform/ board supported Atmega-328 which is of 8-bit, 16 Mhz. during this serial communication is enabled and has 14 digital input /output pins (out of which 6 are PWM) and 6 analog input pins. It operates at 5v. every pin contains a specific function to control.

The storage is non-volatile storage and EEPROM. The key comparison of non-volatile storage with the EEPROM is that the incontrovertible fact that non-volatile storage contents are erasable. In contrast to a EEPROM, the entire device is erased. where one can erase and judge on bases of Byte and section. The availability of the non-volatile memory during which the blocks of the contains are divided and therefore block by block the portion is erased, where an no erased option is provided for the EEPROM byte. thanks to the actual fact that the programming of the non-volatile storage performed while it's on the system board socket. BIOSROM of the PCB is the new upgrade which is immensely used.



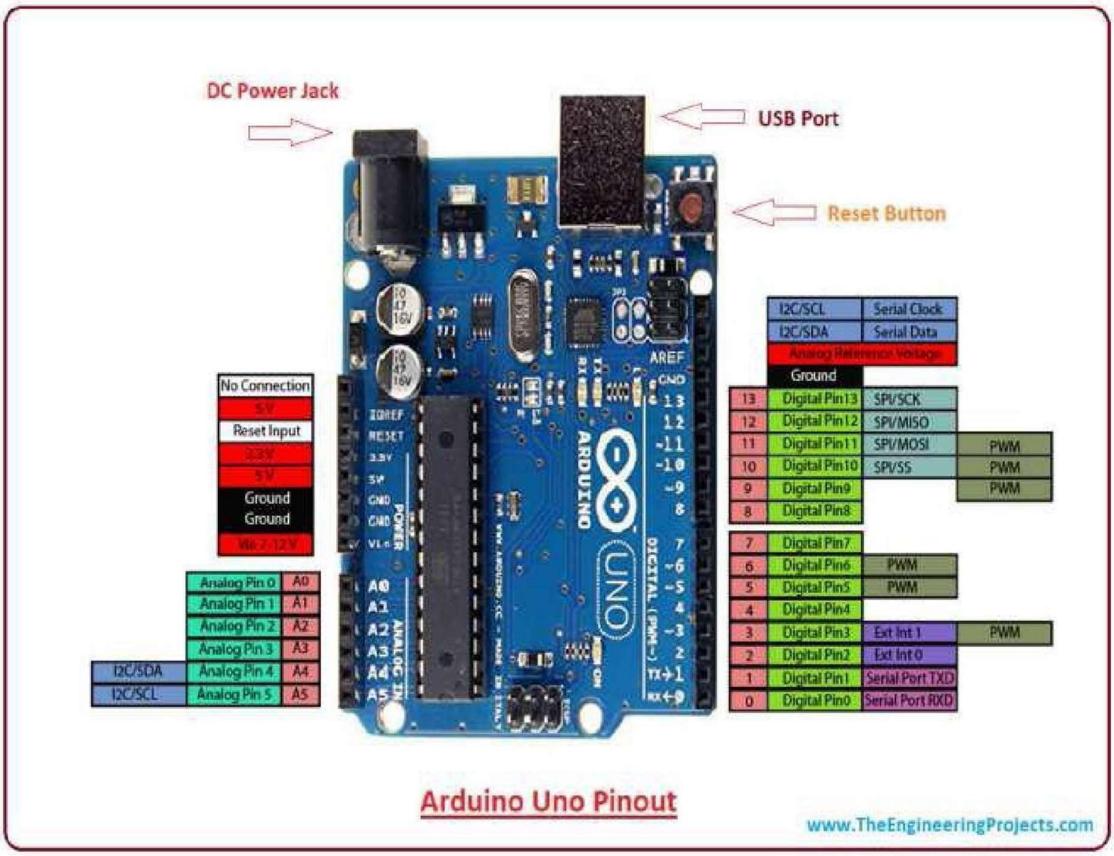
Arduino UNO is based on AVR microcontroller called Atmega328. This controller comes with 2KB SRAM, 32KB of flash memory, 1KB of EEPROM. Arduino Board comes with 14 digital pins and 6 analog pins. ON-chip ADC is used to sample these pins. A 16 MHz frequency crystal oscillator is equipped on the board. Following figure shows the pinout of the Arduino Uno Board.

Pin Description

There are several I/O digital and analog pins placed on the board which operates at 5V. These pins come with standard operating ratings ranging between 20mA to 40mA. Internal pull-up resistors are used in the board that limits the current exceeding from the given operating conditions. However, too much increase in current makes these resisters useless and damages the device.

- LED. Arduino Uno comes with built-in LED which is connected through pin 13. Providing HIGH value to the pin will turn it ON and LOW will turn it OFF.
- Vin. It is the input voltage provided to the Arduino Board. It is different than 5 V supplied through a USB port. This pin is used to supply voltage. If a voltage is provided through power jack, it can be accessed through this pin.
- 5V. This board comes with the ability to provide voltage regulation. 5V pin is used to provide output regulated voltage. The board is powered up using three ways i.e., USB,
- Vin pin of the board or DC power jack. USB supports voltage around 5V while Vin and Power Jack support a voltage range's between 7V to 20V. It is recommended to operate the board on 5V. It is important to note that, if a voltage is supplied through 5V or 3.3V pins, they result in bypassing the voltage regulation that can damage the board if voltage surpasses from its limit.
- GND. These are ground pins. More than one ground pins are provided on the board which can be used as per requirement.
- Reset. This pin is incorporated on the board which resets the program running on the board. Instead of physical reset on the board, IDE comes with a feature of resetting the board through programming.
- IOREF. This pin is very useful for providing voltage reference to the board. A shield is used to read the voltage across this pin which then select the proper power source.

- PWM. PWM is provided by 3,5,6,9,10, 11pins. These pins are configured to provided 8-bit output PWM.
- SPI. It is known as Serial Peripheral Interface. Four pins 10(SS), 11(MOSI), 12(MISO),13(SCK) provide SPI communication with the help of SPI library.
- AREF. It is called Analog Reference. This pin is used for providing a reference voltage to the analog inputs.
- TWI. It is called Two-wire Interface. TWI communication is accessed through Wire Library. A4 and A5 pins are used for this purpose.
- Serial Communication. Serial communication is carried out through two pins called Pin 0 (Rx) and Pin 1 (Tx). Rx pin is used to receive data while Tx pin is used to transmit data.
- External Interrupts. Pin 2 and 3 are used for providing external interrupts. An interrupt is called by providing LOW or changing value.



Communication and Programming

Arduino UNO comes with an ability of interfacing with other other Arduino boards, microcontrollers and computer. The Atmega328 placed on the board provides serial communication using pins like Rx and Tx. The Atmega16U2 incorporated on the board provides a pathway for serial communication using USB com drivers. Serial monitor is provided on the IDE software which is used to send or receive text data from the board. If LEDs placed on the Rx and Tx pins will flash, they indicate the transmission of data. Arduino Uno is programmed using Arduino Software which is a cross-platform application called IDE written in Java. The AVR microcontroller Atmega328 laid out on the base comes with built-in bootloader that sets you free from using a separate burner to upload the program on the board.

MQ135 Sensor:

MQ-135 gas sensor module is shown in fig.3. It is a sensor detector used to detects the flammable gas and smoke concentration of the combustible gas in the air, and output is read in the analog voltage and digital value output. Supply input voltage is 5v. it is very sensitive to H2, LPG, CH4, CO, SMOKE, PROPANE. It has three pins for transmitter, receiver, ground and sensitivity can be adjust by the potentiometer. Detects LPG from 200ppm t

Arduino IDE

The Arduino Integrated Development Environment (IDE) is a cross- platform application written in Java, and derives from the IDE for the Processing programming language and the Wiring projects. It is designed to introduce programming to artists and other newcomers unfamiliar with software development. It includes a code editor with features such as syntax highlighting, brace matching, and automatic indentation, and is also capable of compiling and uploading programs to the board with a single click. A program or code written for Arduino is called a "sketch".

Arduino programs are written in C or C++. The Arduino IDE comes with a software library called "Wiring" from the original Wiring project, which makes many common input/output operations much easier. The users need only to define two functions to make an executable cyclic executive program:

- Setup (): a function run once at the start of a program that can initialize settings
- loop (): a function called repeatedly until the board powers off

A typical first program for a microcontroller simply blinks an LED on and off. In the Arduino environment, the user might write a program like this.

PROJECT SCHEDULING

Scheduling is the allocation of resources

Resources in conceptual sense are time & energy but in practical sense are the time, manpower, equipment applied to material. Scheduling is the process of formalizing the planned functions, assigning the starting and completion dates to each activity which proceeds in a logical sequence and in an orderly and systematic manner. This allows you to see at a glance:

- What the various activities are
- When each activity begins and ends
- How long each activity is scheduled to last
- Where activities overlap with other activities, and by how much
- The start and end date of the whole project

System Analysis

In System Analysis phase, the whole software and hardware development process, the overall software structure and its outlay are defined. The prototype and sketch of the system are designed in analysis phase. Analysis is very important in the whole development cycle process. Any fault in the design phase could be very expensive to solve in the software development process. In this phase, process, the software's overall structure and its nuances are defined. This step refers to the gathering of system requirements, with the goal of determining how these requirements will be accommodated in the system.

Feasibility Study

This study will be bringing the up the existing system and the drawbacks in the existing one, after gathering the information a note of improvements should be done to overcome in our system. The requirements are also considered in the study such as the human effort, resources embedding in the application. In the system we going to propose the resources will be the PDF (should be unique content), videos and PPT and many more. A development team will be required to acquire the goal. Apart from this there are also few requirements will be must and without the system cannot be introduced (Development Requirements) which are hardware and software requirements. The entire life cycle of the development process should be planned and kept on a track using a development cycle to get the idea about the development and the progress we are making. It includes the time planned for each module and its cost.

Preliminary Product Description

Benefit to Organization

The project will be used in industrial as well as household uses for risk free uses of dangerous gases. The Gas Leakage Detection system monitors the environment through the sensors and detects the risk when there is a leakage.

The Initial cost

In manufacturing process, the cost has been emphasized that the expenditure does not exceed out from the expenses scope. Therefore, the importance of survey needs to be done before selection of materials is made. The initial cost of setting up the system will include the cost hardware components, software (server OS, add-on software utilities) and ((setup and maintenance). The same has to nearby the organization

Conceptual Model

Working Principle of the system

This system is based on the Arduino UNO R3 and MQ-135 gas sensor. When the sensor detects gas in the atmosphere, it will give digital output 1 and if gas in not detected the sensor will give digital output 0. Arduino will receive the sensor output as digital input. If the sensor output is high, then the buzzer will start tuning along with the LCD that will show that “Gas detected: Yes”. If the sensor output is low then buzzer will not be tuning, and the LCD will show that “Gas detected: No”. The buzzer most commonly consists of a number of switches or sensors connected to control unit that determines which button was pushed or whether a preset time has lapsed, and usually illuminates a light on the appreciate button or control panel, and sounds a warning in the form of a continuous or intermittent buzzing or beeping sound.

Once the system is installed and power is supplied to the system the Sensor Starts working. The sensor continuously monitors the environment. If the sensor detects the leakage of gas, it transmits a signal to the microcontroller (Arduino). The Arduino board detects the presence of gas concentration in that area. If the gas is in higher concentration and dangerous. The red LED is turned on and the buzzer starts beeping as an alarm for the user, and a message is shown on the LCD display connected to the system. The GSM module also sends an alert message to the registered mobile number. An exhaust fan is also attached to system which reacts when the gas concentration gets high in that area the exhaust fan turned on and throws the gas out from that area.

Prototyping Model

The Prototyping Model is a systems development method in which a prototype is built, tested, and then reworked until an acceptable prototype is finally accomplished from which the complete system or product can now be developed. Thus, this model works best in scenarios where not all of the project requirements are known in detail ahead of time. This model is an iterative, trial-and-error process that takes place bet ween the developers and the users. Other than that, a prototype also is a model or a program which is not based on exact planning but is an early approximation of the final product or software system. There are 4 phases in the prototype model which are requirement gathering, quick design, client evaluation, and refined the end functionality.

The advantages of prototyping model are the users can view their requirements as prototypes as new requirements are being gathered. Besides, the prototyping model has a flexible design and flexibility in development can be achieved and thus a more accurate end product. Other than that, it also can reduce time and costs and of course, there is a great involvement of users in software development. Consequently, the requirements of the users are met to the greatest extent. A systems development life cycle (SDLC) has three primary objectives which are to ensure that high-quality systems are delivered, to provide strong management controls over the projects, and to maximize the productivity of the systems developer.

Phase 1: Requirement Gathering

This is the first phase and there is no planning involved. Instead of that, this application being considered by thinking what is needed for making the genuine product that is going to be developed. This phase is carried out at the earliest stage of each cycle.

The process of gathering requirements is usually more than simply asking the users what they need and writing their answers down and also based on research and review on existing system.

Phase 2: Quick Design

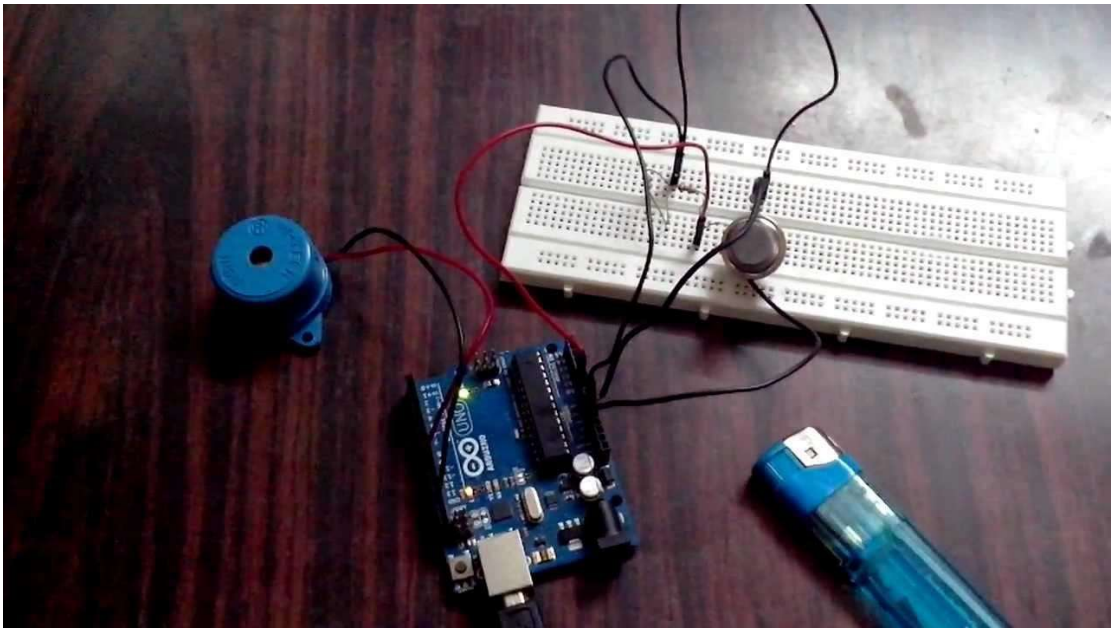
When requirements gathering phase is done, a preliminary design or quick design for the system is created. It is not a detailed design and includes only the prime aspects of the system, which gives an idea of the system to the user. A quick design helps in developing the prototype.

Phase 3: Prototype Evaluated

Prototype evaluated phase is where developer show their prototype to the client and for this proposed project, it is show to the panel. The panel will be evaluated is it satisfy or not.

Phase 4: Refined the End Functionality

This phase is refined the function that cannot be working. First of all, this phase begins with brainstorming session with the supervisor to come up with the solution of the problem that occurred and refined the end functionality for the coding project.



4. SYSTEM DESIGN

Introduction

The gas leakage detector is designed for proper medical attention. Gas cylinder and gas burst is a common thing in our country. Our device can communicate with the nearest hospital and nearest firefighter wirelessly. It can be used in home security, industrial security and vehicle. It can be used in a rural area or highway area for ensuring safety

Our proposed system helps people into three categories:

- Proper fire service station attention
- Provide vital information
- Detects LPG from 200ppm to 10000ppm

Our proposed proposal used:

- IoT based hardware technology
- Scripting programming language

Basic Module

We made this advice based on two main sectors: the transmission and detection module, and receiving module. The detection and transmitting module detect the gas leakage using a sensing circuit built for this purpose. If the sensor detects a gas leakage, it provides an alarm and sends a signal to another module. GSM module will send emergency SMS to fire service station, emergency contacts numbers and power line automatically cutoff. The device was tested using LPG gas and the alarm was activated as a result of gas leakage. However, the former gas leakage system cannot react in time. This project provides the design approach on both software and hardware.

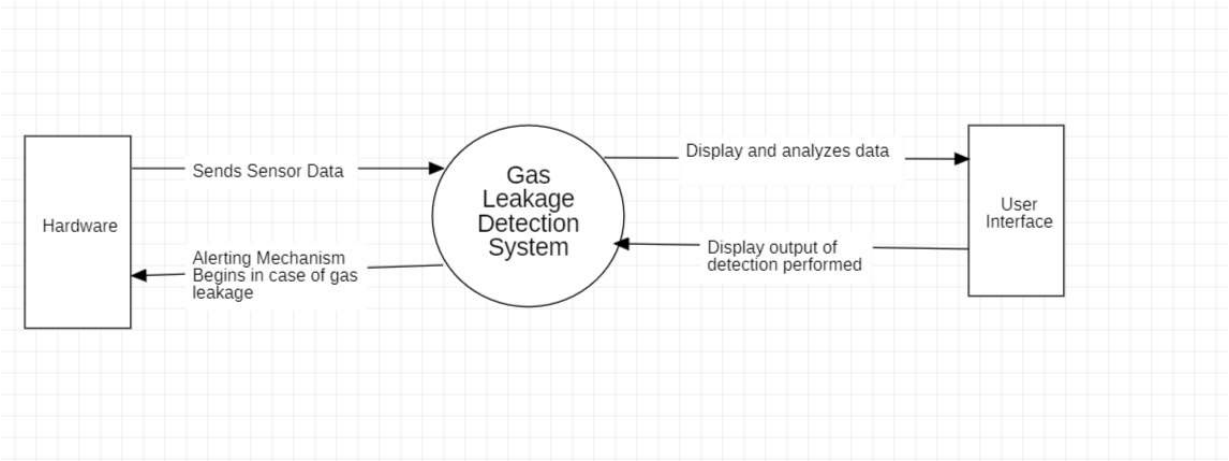
DATA DESIGN

Data Flow Diagram

A data flow diagram (DFD) maps out the flow of information for any process or system. It uses defined symbols like rectangles, circles and arrows, plus short text labels, to show data inputs, outputs, storage points and the routes between each destination. Data flowcharts can range from simple, even hand-drawn process overviews, to in-depth, multi-level DFDs that dig progressively deeper into how the data is handled.

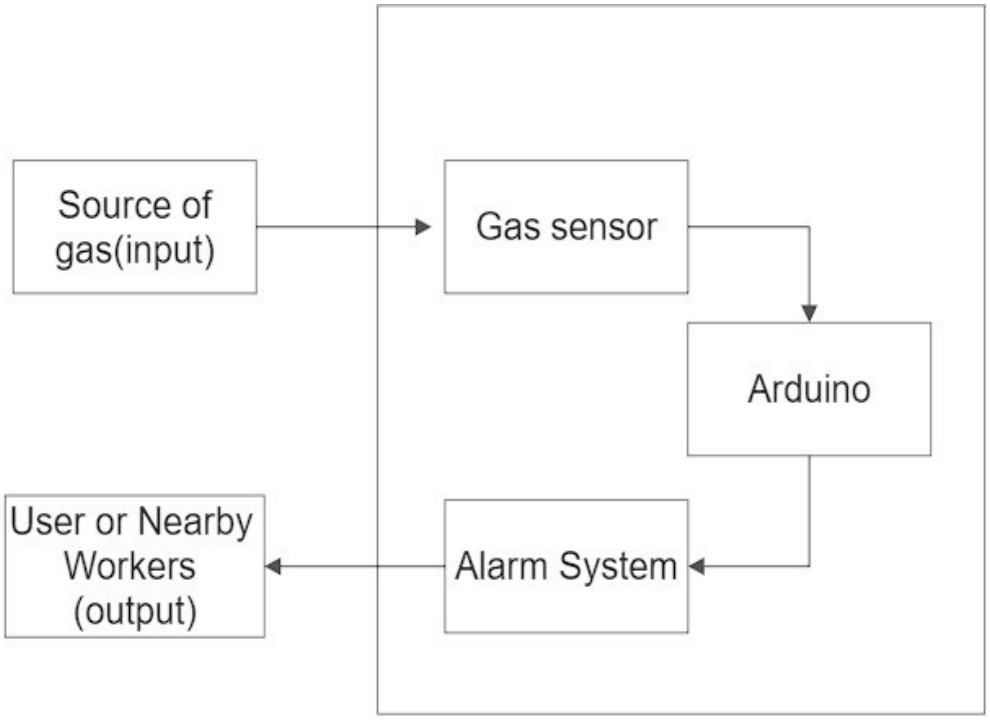
- Each process should have at least one input and an output.
- Each data store should have at least one data flow in and one data flow out.
- Data stored in a system must go through a process.
- All processes in a DFD go to another process or a data store.

Data Flow diagram



Block Diagram of Gas Leakage Detector

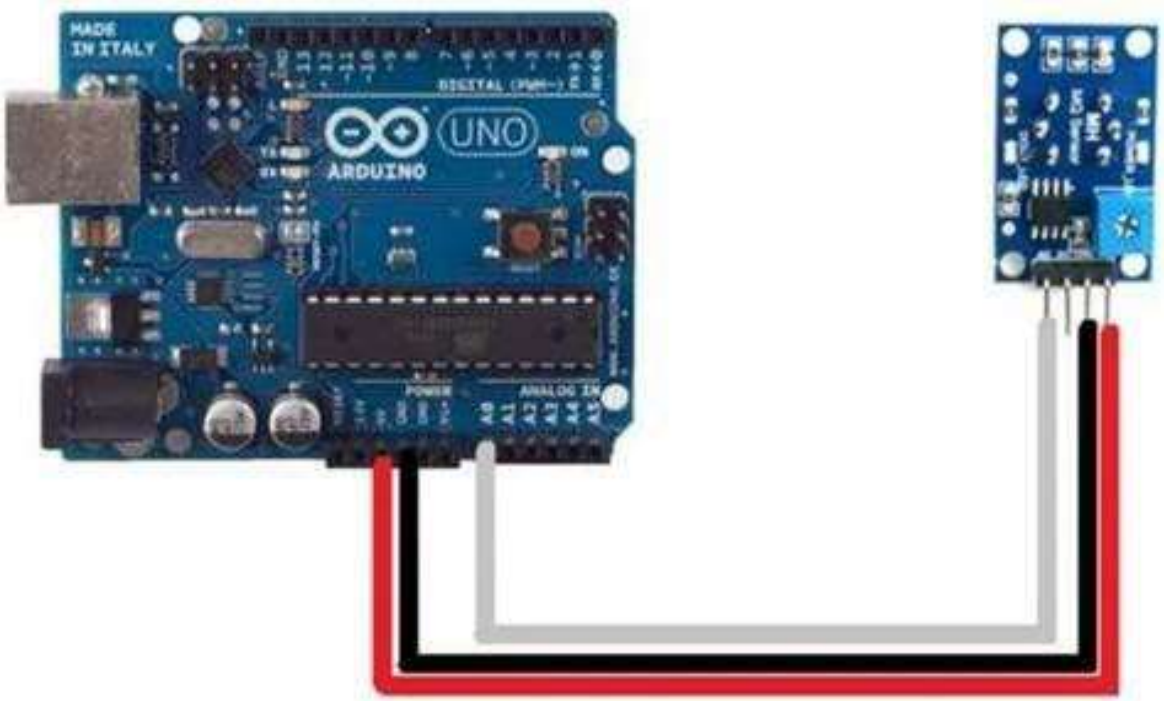
In today's era especially in the women, the craze of gas leakage detector is really remarkable. As the gas cylinder burst accident was increasing, due to which many deaths occur, most of them are caused not taking any security to detect the accident and also due to the poor medical facility. So, providing safety to a person while gas leakage detector is a prime concern. Thus, by considering this problem, the gas leakage detector gives them necessary safety and they can easily understand the gas leakage detector is essential needs to their home safety, if not the system won't work.



Safety Helmet for Gas Leakage

Final Circuit Diagram

In our project we have used various components in the IOT and Arduino based Gas Leakage detection system. MQ gas sensor is used to detect the gas leakage.

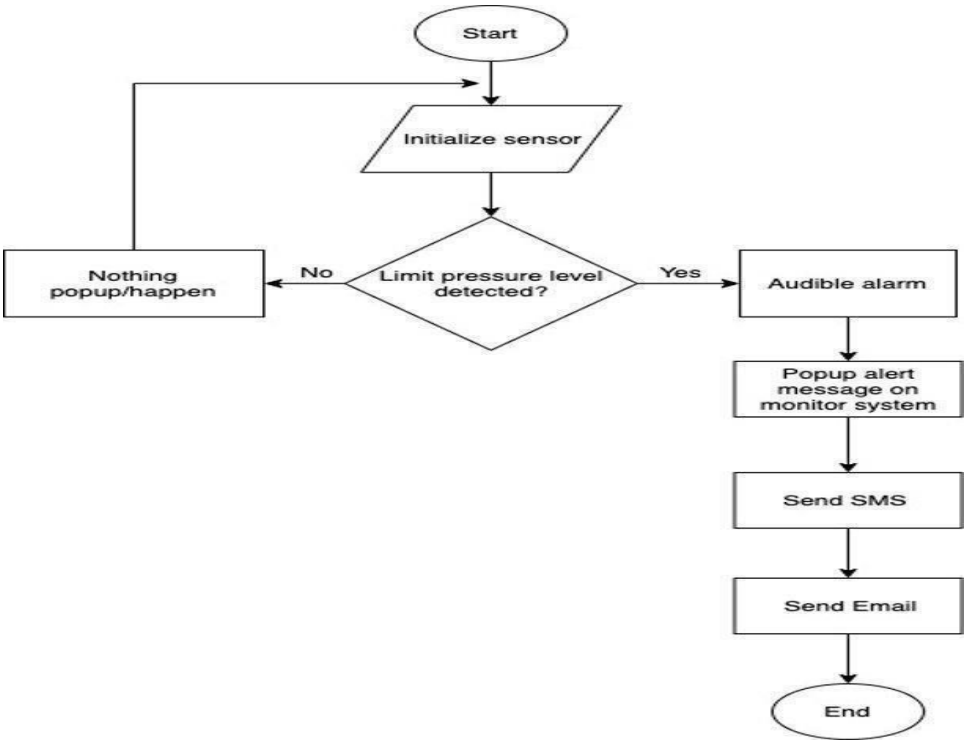


Arduino is used to turn on buzzer, send a message to the LCD and send data to the GSM module which sends the alert message to the user. LCD is used to display an informative message. A buzzer is used to signal the gas leakage. The below diagram shows the final circuit diagram and the connections done in the circuit.

As shown in the schematic diagram above. It contains Arduino board. Gas sensor Module, buzzer and 16 x2 LCD module. Arduino controls the whole process of this System like reading Gas Sensor Module output, sending message to LCD and active ting buzzer. We can set sensitivity of this sensor module by inbuilt potentiometer placed on it.

Gas sensor module DO pin is directly connected to pin 18(A4) of Arduino and VCC and GND are connected to VCC and GND of Arduino. LPG Gas sensor module consists some heater supply to heat up and it may take up to 15 minutes to get ready for detecting Gas. And a comparator circuit is used for converting Analog output of MQ sensor in digital directly connected to Arduino pin 2, GND and 3. And data pin D0-D7 are connected to 4,5,6,7 of Arduino. A buzzer is connected with Arduino pin number 13 through a NPN BC547 transistor having a 1k resistor at its base.

Algorithm



DATA INTEGRITY AND CONSTRAINTS

Data integrity is the overall accuracy, completeness, and consistency of data. Data integrity also refers to the safety of data in regards to regulatory compliance and security. It is maintained by a collection of processes, rules, and standards implemented during the design phase. When the integrity of data is secure, the information stored in a database will remain complete, accurate, and reliable no matter how long it's **stored** or how often it's accessed. Data integrity also ensures that your data is safe from any outside forces. Integrity constraints are a set of rules. It is used to maintain the quality of information. Integrity constraints ensure that the data insertion, updating, and other processes have to be performed in such a way that data integrity is not affected.

SECURITY ISSUES

Security basically is the protection of computer systems or designed systems and information from harm, theft, and unauthorized use. It is the process of preventing and detecting unauthorized use of your computer system. Cyber security is defined as protecting computer systems, which communicate over the computer networks. Security is important because it keeps your information protected. It's also important for your computers or for system's overall health; proper computer security helps prevent viruses and malware, which allows programs to run quicker and smoother. Security in the system helps the system to tackle some infections such as Trojan, Worms, and Viruses etc.

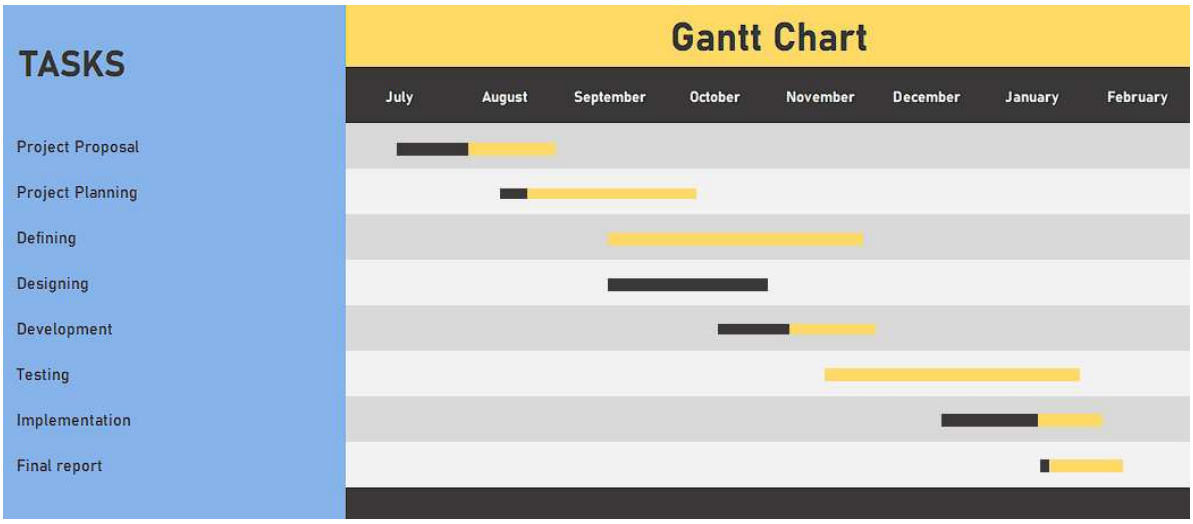
Testing Period

Testing was carried out by releasing LPG into the atmosphere around the sensor. The gas detector and response unit with gas lighter used for the test and test setup respectively.

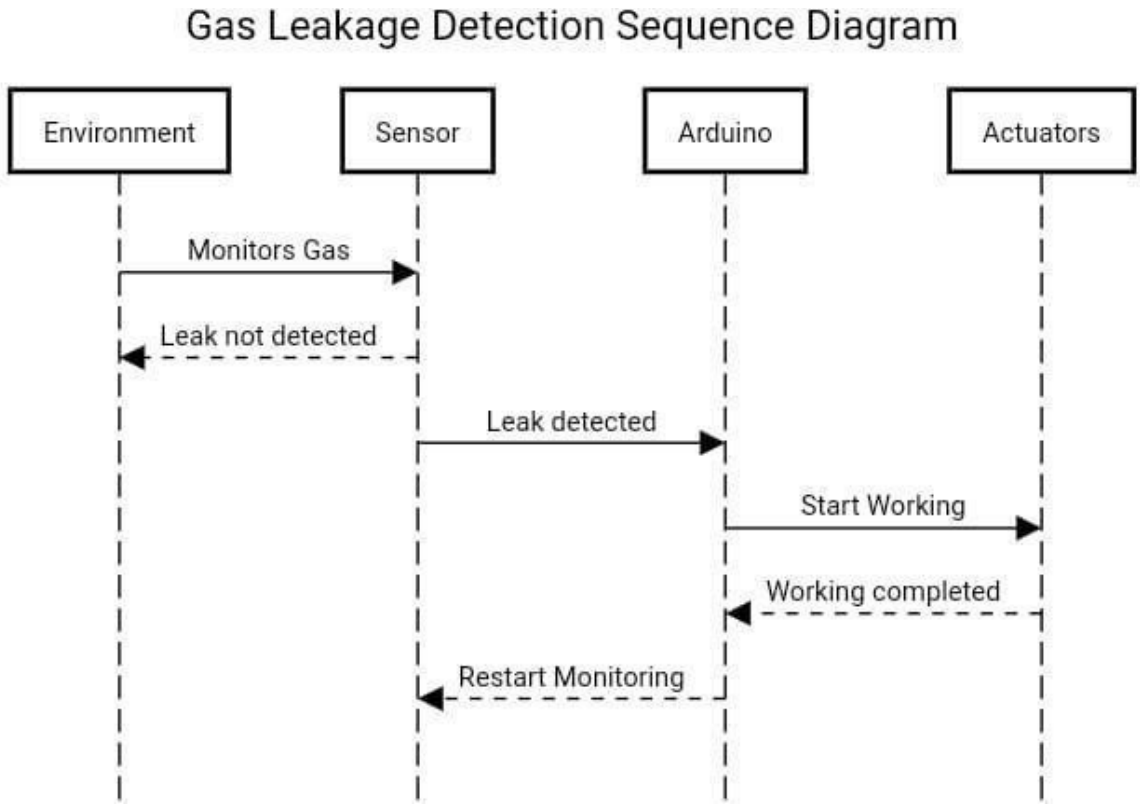
It was observed that when the LPG device was tested by placing it at different distances from the gas source, the response time of the LPG system decreased as the distance from the gas source increased and vice versa. Also, it was observed that the sensitivity of the gas sensor was very high in clean air. The gas sensor' sensitivity varied with temperature/Humidity while the reference voltage remained constant over time with error of +1v. At constant gas concentration, the sensed voltage will always be constant. The gas sensor has a very fast response to gas since the time difference between test results with same concentration is very small while the difference between the sensed voltages is very high.

GANTT CHART

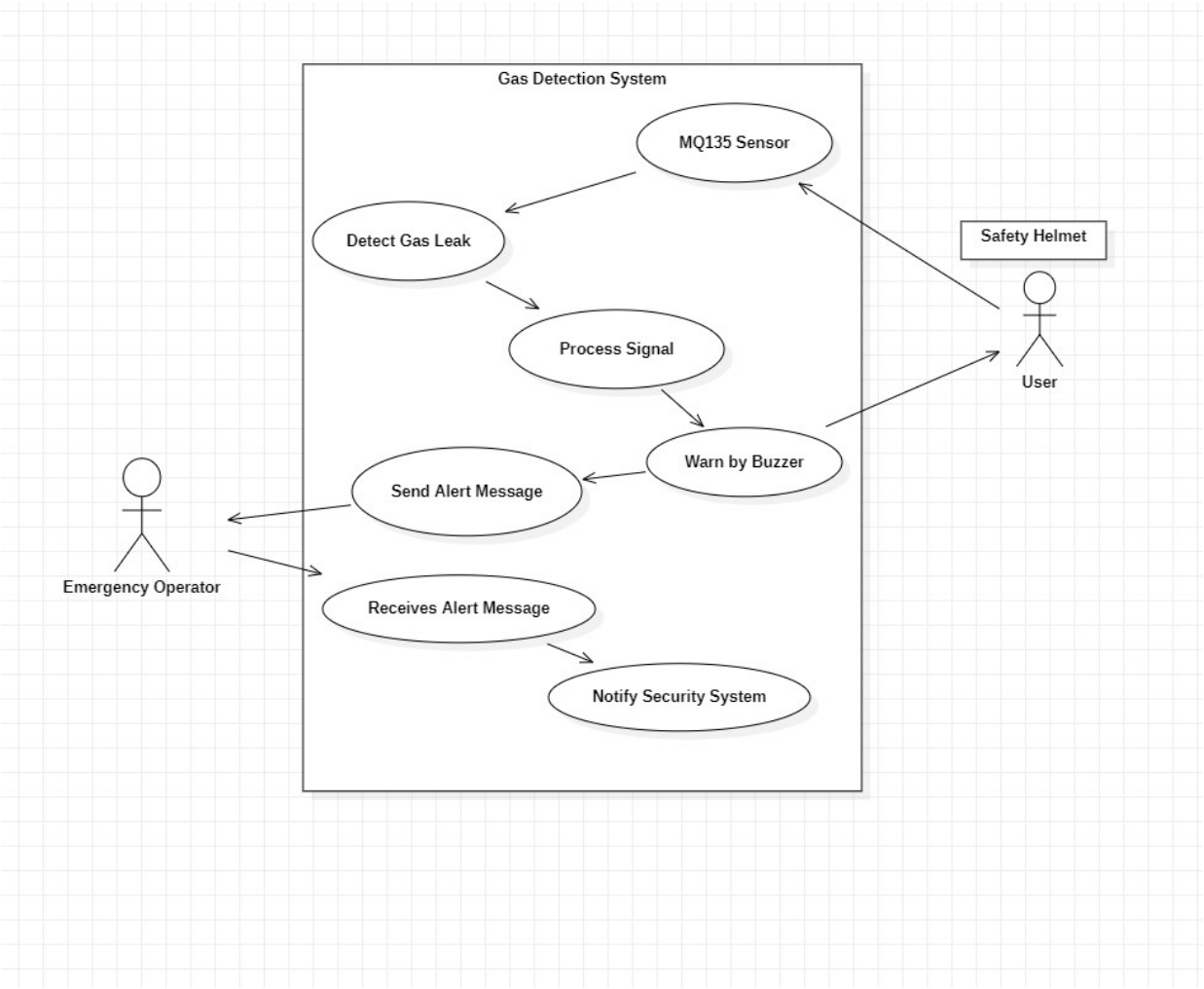
Gantt chart are a used in project development lifecycle which shows the complete process of project development by time period required for project development. project scheduling is very important to keep in mind the progress of the project according to the time required. Gantt Chart defines all the process of development and specifies the time according to the time period.



Sequence Diagram



Use Case Diagram



Summary

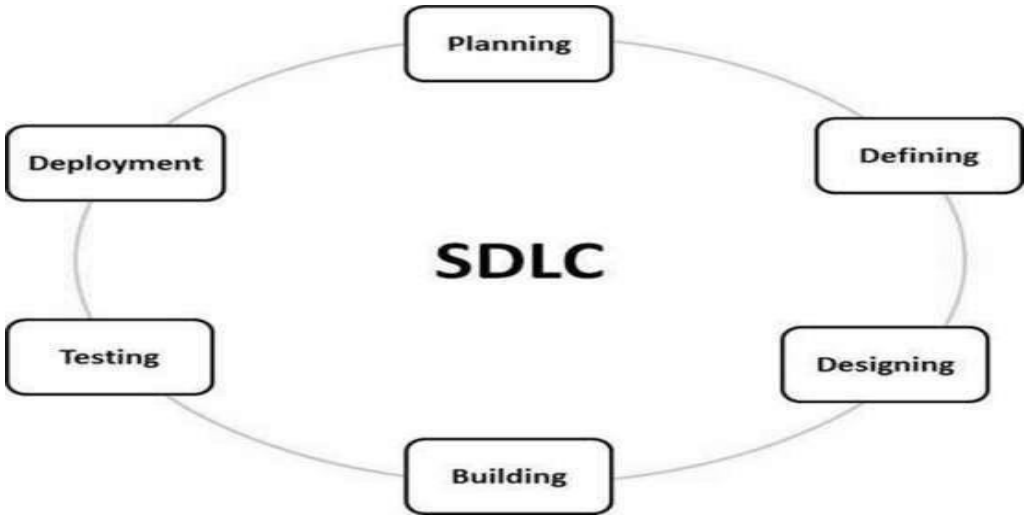
"Safety Helmet for Gas leakage" is a project based on to enhance man and machine safety. In a petrochemical industry. As petroleum industries are the largest process control industry it is also highly prone to major fire and gas disasters. A chemical industry has excessively high amount of crude oil stored within a confined area. Therefore, presence of any external source which can cause heat or fire would lead to a major disaster. Even the gas that are present in petroleum refineries are hazardous. And another instance, the Vishakhapatnam, HPCL refinery tragedy claimed lives of 30people. Though a gas and fire detection system were present which is connected to the sensors using large number of wires that run from the control room to various plant areas. But during the fire the wire itself got damaged, so the information did not reach the control room. Therefore, our system is developed with the aim of overcoming the restrictions and disadvantages of the existing system. The system we have designed is an integrated system which will monitor timely gas leakage in any area around the plant using Arduino which is a wireless communication device. We have also proposed a new system which monitors human density within the plant area. Therefore, Integrated plant safety monitor system based on Arduino can realize workers attendance registration, Real-time precise positioning, Dynamic gas concentration monitoring, Real-time data transmission & Danger alarm.

5. IMPLEMENTATION AND TESTING

IMPLEMENTATION APPROACHES

Detailed Life Cycle of the Project

Making an IoT is no mystery in today's time however making successful IoT is a process which involves quite an extensive pre-planning. There has been a meteoric rise in the IoT market in the past few years with millions of IoTs in the two prominent IoT stores. In such a case when the competition is so huge, also you have to really make it stand out in function, in its use and in its design to attract the consumer's attention. And to achieve all this you have to get it right the very first time. Since, it is so important for an IoT to be spot-on the first time hence it is necessary for developers to follow a step-by-step process to building an IoT. We build a mobile IoT in different phases and each phase has a number of steps, involved. To understand it further – Internet of Things Lifecycle is just a representation of the conventional Software Development Lifecycle (SDLC) but from the perspective of a mobile device. So, let's see the development phases.



Phases:

Planning

Planning phase is used to gather the requirements which will be used to develop the system, its need in the society and what the service the system is providing.

Designing

The IoT design approach is an approach to the IoT system development with respect to the peculiarities of the Internet of Things. That term covers all components of IoT architecture from IoT devices and their hardware to applications and user interfaces. The term Internet of Things design includes shapes of devices, their material, colour scheme, a cloud service provider, modules, applications, and much more.

Building

Building IoT refers to Internet of Things sensors specifically designed to enhance the occupant experience and operations of the built environment.

Defining

This phase will co relate the entire information and the resources which we have collected and its uses in our development phase, where our system will use the information-based formats like PDF, videos and many more. Designing A prototype of the system will be designed exactly similar to the real-time system we are going to develop i.e., its UI, the page format and the look and feel of the application. Building The actual development (coding) according to the requirement is done in this phase, which will include building the Xml design page and Java handling page in our case (building of the modules).

Testing

All the modules designed will be checked and tested to see if the desired output is acquired from the built module.

Deployment

This is the last phase of the development in which the system is handover

Coding Details and Code Efficiency

Coding Details

Coding is the process by which an individual researcher takes a piece of information from legislation, policy, or any other source and translates it into a set of characteristics that can be quantitatively analyzed. Coding is the process of using a programming language to get a computer to behave how you want it to. Every line of code tells the computer to do something, and a document full of lines of code is called a script. Each script is designed to carry out a job. This job might be to take an image and change its size.

```
int gas;

int x;

int a=0,b=0;

void setup() {

  pinMode(13, OUTPUT);

pinMode(10, OUTPUT);

pinMode(4, OUTPUT);

pinMode(RELAY,OUTPUT);

pinMode(sensor,INPUT);

pinMode(buzzer,OUTPUT);

Serial.begin(9600);

delay(2000);

  Serial.println("AT+CMGF=1");

  delay(2000);
```

Code Efficiency

Code efficiency is a broad term used to depict the reliability, speed and programming methodology used in developing codes for an application. Code efficiency is directly linked with algorithmic efficiency and the speed of runtime execution for software. It is the key element in ensuring high performance. The goal of code efficiency is to reduce resource consumption and completion time as much as possible with minimum risk to the business or operating environment. The software product quality can be accessed and evaluated with the help of the efficiency of the code used.

Code efficiency plays a significant role in applications in a high execution- speed environment where performance and scalability are paramount.

One of the recommended best practices in coding is to ensure good code efficiency. Well-developed programming codes should be able to handle complex algorithms.

Recommendations for code efficiency include

- To remove unnecessary code or code that goes to redundant processing
- To make use of optimal memory and non-volatile storage
- To ensure the best speed or run time for completing the algorithm
- To make use of reusable components wherever possible
- To make use of error and exception handling at all layers of software, such as the user interface, logic and data flow
- To create programming code that ensures data integrity and consistency
- To develop programming code that's compliant with the design logic and flow
- To make use of coding practices applicable to the related software
- To optimize the use of data access and data management practices
- To use the best keywords, data types and variables, and other available programming concepts to implement the related algorithm.
- Code efficiency is a broad term used to depict the reliability, speed and programming methodology used in developing codes for an application.
- Code efficiency is directly linked with algorithmic efficiency and the speed of runtime execution for software.

- It is the key element in ensuring high performance
- The software product quality can be accessed and evaluated with the help of the efficiency of the code used.

MVC Architecture

The Model-View-Controller (MVC) framework is an architectural pattern that separates an application into three main logical components Model, View, and Controller. Hence the abbreviation MVC. Each architecture component is built to handle specific development aspect of an application. MVC separates the business logic and presentation layer from each other. It was traditionally used for desktop graphical user interfaces (GUIs). Nowadays, MVC architecture has become popular for designing web applications as well as mobile apps.

Features of MVC

- Easy and frictionless testability. Highly testable, extensible and pluggable framework
- Offers full control over your HTML as well as your URLs
- Leverage existing features provided by ASP.NET, JSP, Django, etc.
- Clear separation of logic: Model, View, Controller. Separation of application tasks viz. business logic, Ul logic, and input logic
- URL Routing for SEO Friendly URLs. Powerful URL- mapping for comprehensible and searchable URLs

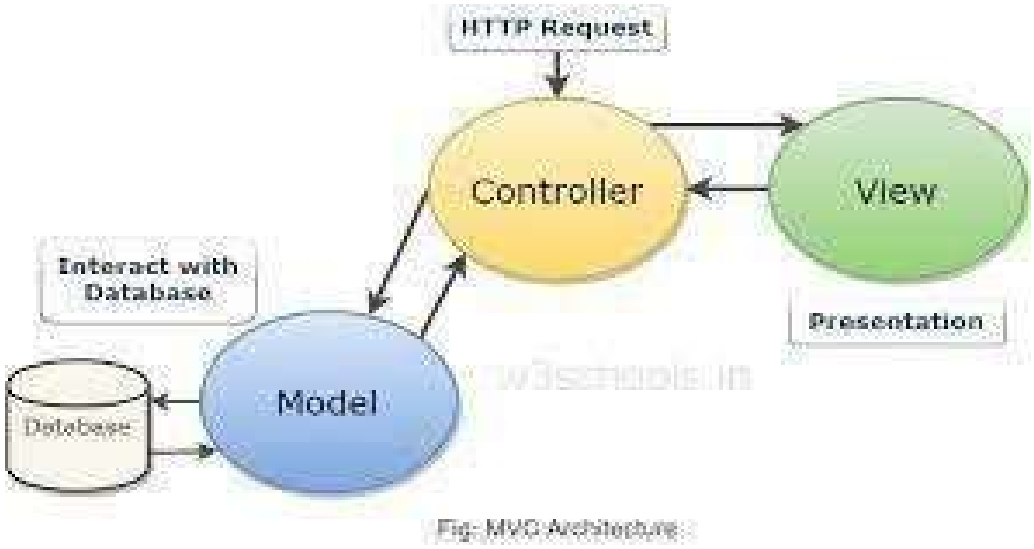


Fig: MVC Architecture

Three Important MVC components are:

- Model: It includes all the data and its related logic
- View: Present data to the user or handles user interaction
- Controller: An interface between Model and View components

Let's see each other this component in detail:

View:

A View is that part of the application that represents the presentation of data. Views are created by the data collected from the model data. A view requests the model to give information so that it resents the output presentation to the user.

The view also represents the data from chats, diagrams, and table. For example, any customer view will include all the UI components like text boxes, drop downs, etc. The view in our system is the GUI interface of the application which interacts with the user when they use the application. View acts as a guide for the user and allows accessing the services of the system.

Controller:

The Controller is that part of the application that handles the user interaction. The controller interprets the mouse and keyboard inputs from the user, informing model and the view to change as appropriate.

A Controller sends commands to the model to update its state (E.g., Saving a specific document). The controller also sends commands to its associated view to change the view's presentation (For example scrolling a particular document). Controller in other words can be the database or the server which accepts the requests from the system and responses accordingly. In our case firebase is the main controller in our system which takes and holds the data which is provided in the front end.

Model:

The model component stores data and its related logic. It represents data that is being transferred between controller components or any other related business logic. For example, a Controller object will retrieve the customer info from the database. It manipulates data and sends back to the database or uses it to render the same data. It responds to the request from the views and also responds to instructions from the controller to update it. It is also the lowest level of the pattern which is responsible for maintaining data. The uploadation of videos, retrieval of e-books, replying to the answers etc. comes under the business logics that are the services of the system.

TESTING APPROACH

Software Testing has a dual function; it is used to identify the defects in program and it is used to help judge whether or not program is usable in practice. Thus, software testing is used for validation and verification, which ensure that software conforms to its specification and meets need of the software customer. Developer resorted Alpha testing, which usually comes in after the basic design of the program has been completed. The project scientist will look over the program and give suggestions and ideas to improve or correct the design. They also report and give ideas to get rid of around any major problems. There is bound to be a number of bugs after a program have been created. Tests are grouped together based on where they are added in SDLC or the by the level of detailing they contain. In general, there are four levels of testing: unit testing, integration testing, system testing, and acceptance testing. The purpose of Levels of testing is to make software testing systematic and easily identify all possible test cases at a particular level. There are many different testing levels which help to check behavior and performance for software testing. These testing levels are designed to recognize missing areas and reconciliation between the development lifecycle states. In SDLC models there are characterized phases such as requirement gathering, analysis, design, coding or execution, testing, and deployment.

The Testing Process

We test the software process activities such as Design, Implementation, and Requirement Engineering. Because, design errors are very costly to repair once system has been started to operate, it is quite obvious to repair them at early stage of the system. So, analysis is the most important process of any project.

Requirement Traceability

As most interested portion is whether the system is meeting its requirements or not, for that testing should be planned so that all requirements are individually tested. We checked the output of certain combination of inputs so that we can know whether it gives desirable results or not. Strictly sticking to your requirements specifications, give you the path to get desirable results from the system.

Testing Schedule

We have tested each procedure back-to-back so that errors and omissions can be found as early as possible. Once the system has been developed fully, we tested it on other machines, which differs in configuration.

Testing Strategy

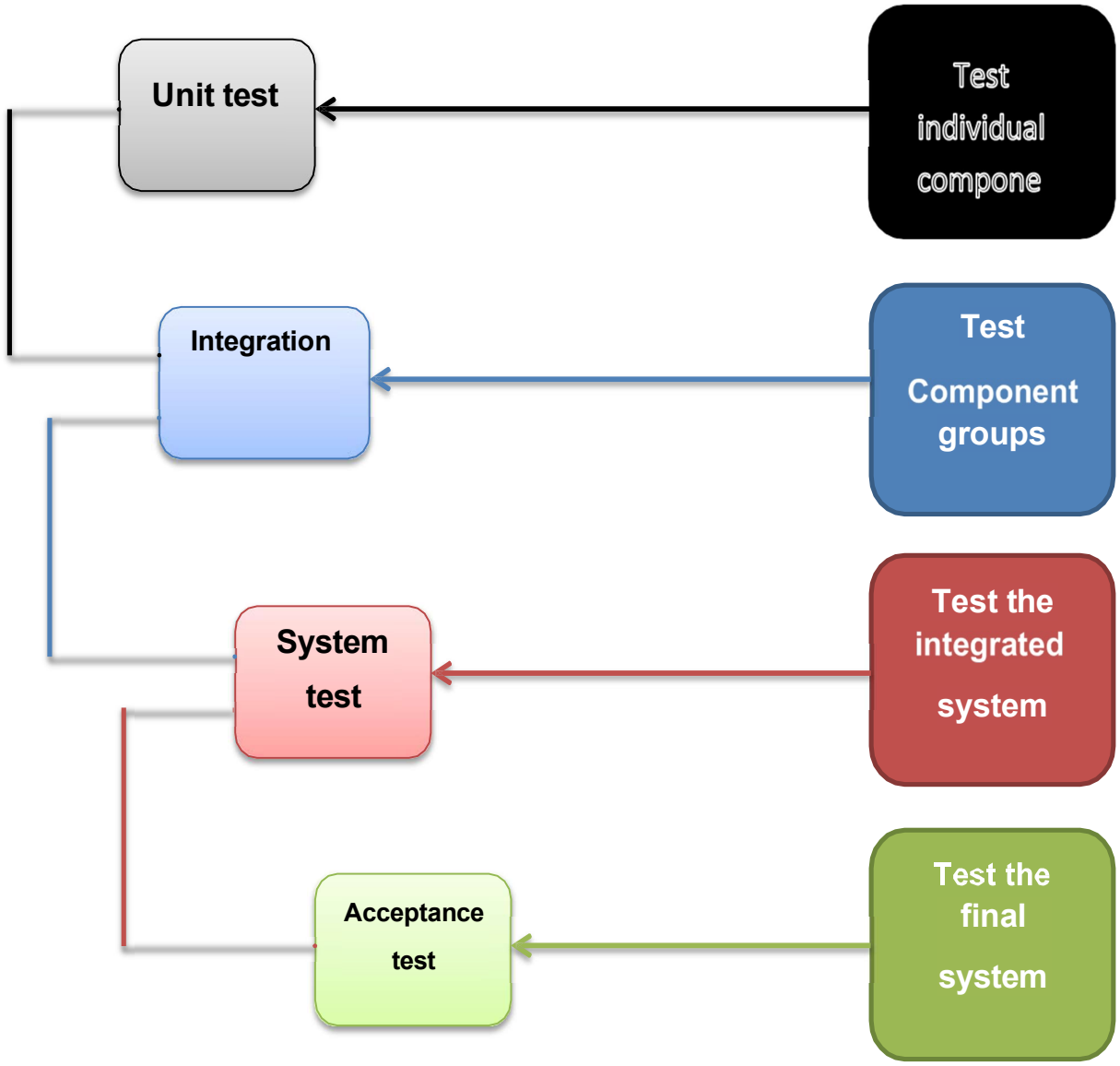
There are types of testing that we implement. They are as follows:

- While deciding on the focus of testing activities, study project priorities. For example, for an on-line system, pay more attention to response time. Spend more time on the features used frequently.
- Decide on the effort required for testing based on the usage of the system. If the system is to be used by a large number of users, evaluate the impact on users due to a system failure before deciding on the effort.
- A necessary part of the test case is a definition of the expected result.
- Write test cases for invalid and unexpected as well as valid and expected input conditions.
- Thoroughly inspect the results of each test

All these phases go through the process of software testing levels. There are mainly four testing levels are:

1. Unit Testing
2. Integration Testing
3. System Testing
4. Acceptance Testing

Each of these testing levels has a specific purpose. These testing level provide value to the software development lifecycle.


Unit test
Test individual compone
Integration
Test Component groups
System test
Test the integrated system
Acceptance test
Test the final system

1) Unit Testing:

A Unit is a smallest testable portion of system or application which can be compiled, liked, loaded, and executed. This kind of testing helps to test each module separately. The aim is to test each part of the software by separating it. It checks those components are fulfilling functionalities or not. This kind of testing is performed by developers. Each unit of the modules like the fetching data from the textboxes, cropping images, retrieving images are tested number of times before building into module.

2) Integration Testing:

Integration means combining. For Example, In this testing phase, different software modules are combined and tested as a group to make sure that integrated system is ready for system testing. Integrating testing checks the data flow from one module to other modules. This kind of testing is performed by testers. Modules which work dependently on each other like register module and login module are tested.

3) System Testing:

System testing is performed on a complete, integrated system. It allows checking system's compliance as per the requirements. It tests the overall interaction of components. It involves load, performance, reliability and security testing. System testing most often the final test to verify that the system meets the specification. It evaluates both functional and non- functional need for the testing. The overall system has been checking till date to find the efficiency as well as service problems.

4) Acceptance Testing:

Acceptance testing is a test conducted to find if the requirements of a specification or contract are met as per its delivery. Acceptance testing is basically done by the user or customer. However, other stockholders can be involved in this process.

Other Types of Testing:

- Regression Testing
- Buddy Testing
- Alpha Testing
- Beta Testing

Beta Testing:

Beta Testing is one of the Acceptance Testing types, which adds value to the product as the end-user (intended real user) validates the product for functionality, usability, reliability, and compatibility. Inputs provided by the end-users helps in enhancing the quality of the product further and leads to its success. This also helps in decision making to invest further in the future products or the same product for improvisation. Since Beta Testing happens at the end user's side, it cannot be the controlled activity.

Beta Testing is one of the Customer Validation methodologies to evaluate the level of customer satisfaction with the product by letting it to be validated by the end users, who actually use it, for over a period of time. Product experiences gained by the end users are asked for feedback on design, functionality, and usability and this helps in assessing the quality of the product. Real People, Real Environment, Real Product are the three Rs of Beta Testing and the question that arises here in Beta Testing is "Do Customers like the Product?"

MODIFICATIONS AND IMPROVEMENTS

This project has been improved from the existing gas detector. However, there are some improvements can be made to achieve a better result. Thus, it will make the project even better than the previous gas detector and more efficient. Gas Leakage Detector with Notifier System is the innovation from existing Gas Detector. This project is mainly for an individual who wants to keep their home safe from the risk of burning. The benefit of this project is to detect any leakage of cooking gas and it will alert the user about leakage.

As discussed in previous research work, it is clear that the result is not satisfactory and further research must be done. This project can be improved in future. This is because SMS NOTIFER GAS LEAKAGE DETECTOR only can detect the gas and notify it to the user about the leakage but has not safety to extinguish the fire if the fire occurs. It also cannot reduce the concentration of LPG. When the issues are obvious, so improvement must be made by suggesting to install sprinkle.

Sprinkle is for in the case fire occur. Other than that, there is an issue where gas detector is an electrical device that is dangerous when it is installed near to the LPG barrel. If electric shock occur it will cause electric spark that can cause fire. The wires are messy and too long to connect to other components such as Arduino UNO, gas sensor, buzzer and GSM Module. So, for the betterment of the project, it is encouraged to use Bluetooth. The sensor will stick to the LPG barrel by using magnet.

For improvement in the future, some additional feature could be added in order to make increase the performance and capability of the circuit. Some recommendations are briefly explained below: 5.2.1 Various Gas Detection This model can only detect various combustible gasses at certain distance. By applying the theory of infrared absorption, gassed like Sulphur, carbon Monoxide, Nitrogen Dioxide and other gasses which are toxic and poisonous can only be detected by infrared at certain wavelength. The future improvement of this project will help to improve the functionality of this model 5.2.2 Performance of the circuit When the LPG has dispersed into the surrounding, its concentration will be decreased. Therefore, this model will not be able to detect the leakage appropriately. The design of this model has to be improved so that it can detect the leakage just as the leakage had occurred.

Test Cases

- A test case is a set of conditions or variables and inputs that are developed for a particular goal or objective to be achieved on a certain application to judge its capabilities or features.
- It might take more than one test case to determine the true functionality of the application being tested. Every requirement or objective to be achieved needs at least one test case. Some software development methodologies like Rational Unified Process (RUP) recommend creating at least two test cases for each requirement or objective; one for performing testing through positive perspective and the other through negative perspective.

Test Case Structure

Testing Period Testing was carried out by releasing LPG into the atmosphere around the sensor. The gas detector and response unit with gas lighter used for the test and test setup respectively. It was observed that when the LPG device was tested by placing it at different distances from the gas source, the response time of the LPG system decreased as the distance from the gas source increased and vice versa. Also, it was observed that the sensitivity of the gas sensor was very high in clean air. The gas sensor' sensitivity varied with temperature/Humidity while the reference voltage remained constant over time with error of +1v. At constant gas concentration, the sensed voltage will always be constant. The gas sensor has a very fast response to gas since the time difference between test results with same concentration is very small while the difference between the sensed voltages is very high.

Results

The results of test carried out on the device at different time and day for concentration of gas in the air around the sensor is shown in Table 4.1. The last four values are the case of an endless loop due to high gas concentration and was carried out on 25/01/2023.

The device was tested placing the LPG device at different distances from the gas source.

Set Point voltage	Sensed Voltage	Gas Leakage Detected?	Time
111	97	No	4.19p.m
111	111	Yes	4.20p.m
110	78	No	11.41.29a.m
110	78	No	11.41.35a.m
110	78	No	5.53p.m
110	78	No	5.53.37p.m
110	79	No	5.54.03p.m
110	90	No	5.54.48 p.m
110	84	No	5.55.02 p.m
111	82	No	5.55.59 p.m
110	81	No	5.56.18 p.m
110	81	No	5.56.26 p.m

Fig: Test Results

Designing

Test Cases Test cases should be designed and written by someone who understands the function or technology being tested. A test case should include the following information

- Purpose of the test
- Software requirements and Hardware requirements (if any)
- Specific setup or configuration requirements
- Description on how to perform the test(s)
- Expected results or success criteria for the test

Designing test cases can be time consuming in a testing schedule, but they are worth giving time because they can really avoid unnecessary retesting or debugging or at least lower it. Organizations can take the test cases approach in their own context and according to their own perspectives. Some follow a general step way approach while others may opt for a more detailed and complex approach. It is very important for you to decide between the two extremes and judge on what would work the best for you. Designing proper test cases is very vital for your software testing plans as a lot of bugs, ambiguities, inconsistencies and slip ups can be recovered in time as also it helps in saving your time on continuous debugging and re-testing test cases.

Test Case Content

A test case should include

1. The purpose and conditions for its execution,
2. A step-by-step setup of the test environment,
3. Input data,
4. The expected result,
5. The actual result,
6. A version description of the software,
7. The software's working environment, and
8. The test ID.

6. RESULTS AND DISCUSSION

RESULT

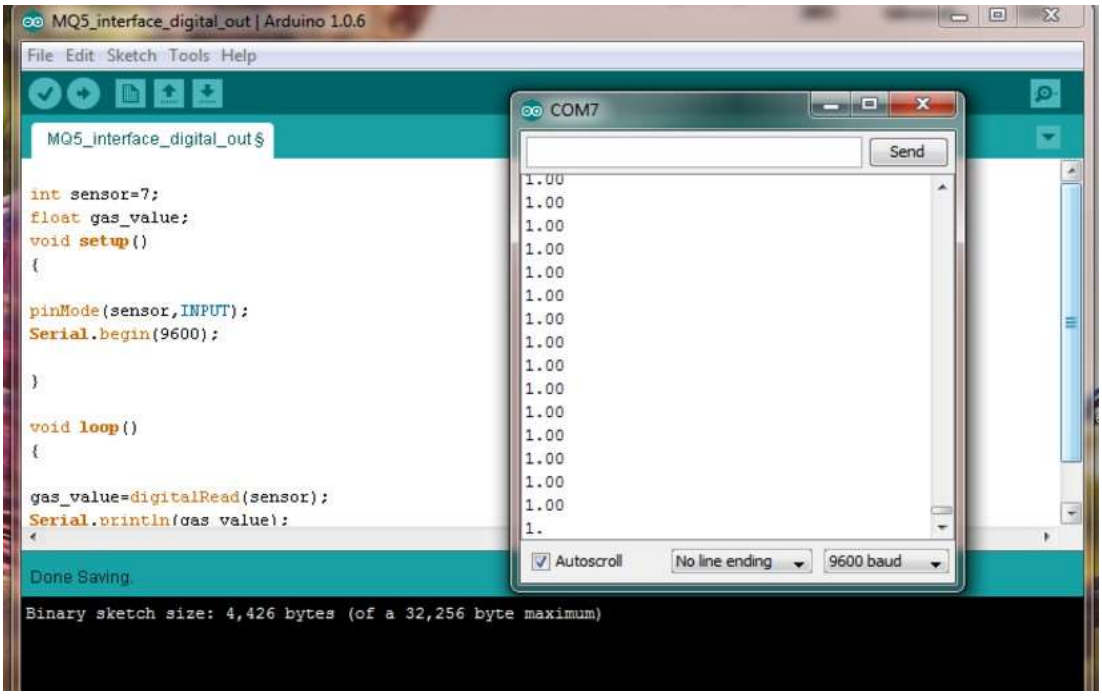
The working of LPG leakage detection system depends on MQ2 sensor. The purpose of sensor is to detect any inflammable gas near its vicinity. The buzzer will work only when sensor does its job properly. The performance of LPG leakage detection system is depending on how well sensor performs. This makes MQ2 sensor crucial component of the present mechatronic system. Hence, the performance of sensor should be tested under various conditions to make sure sensor will perform under different working conditions.

LPG leakage detection system is designed to be used for domestic purpose. The system should operate under all temperature conditions throughout the day. The sensor will operate when inflammable gases will come in vicinity. Hence, the distance between LPG leakage detection system and leakage site plays important role in system operation.

An experiment is carried out to test the performance of sensor at various temperature conditions throughout day. Another experiment is carried out to test performance of sensor with respect to its distance from leakage.

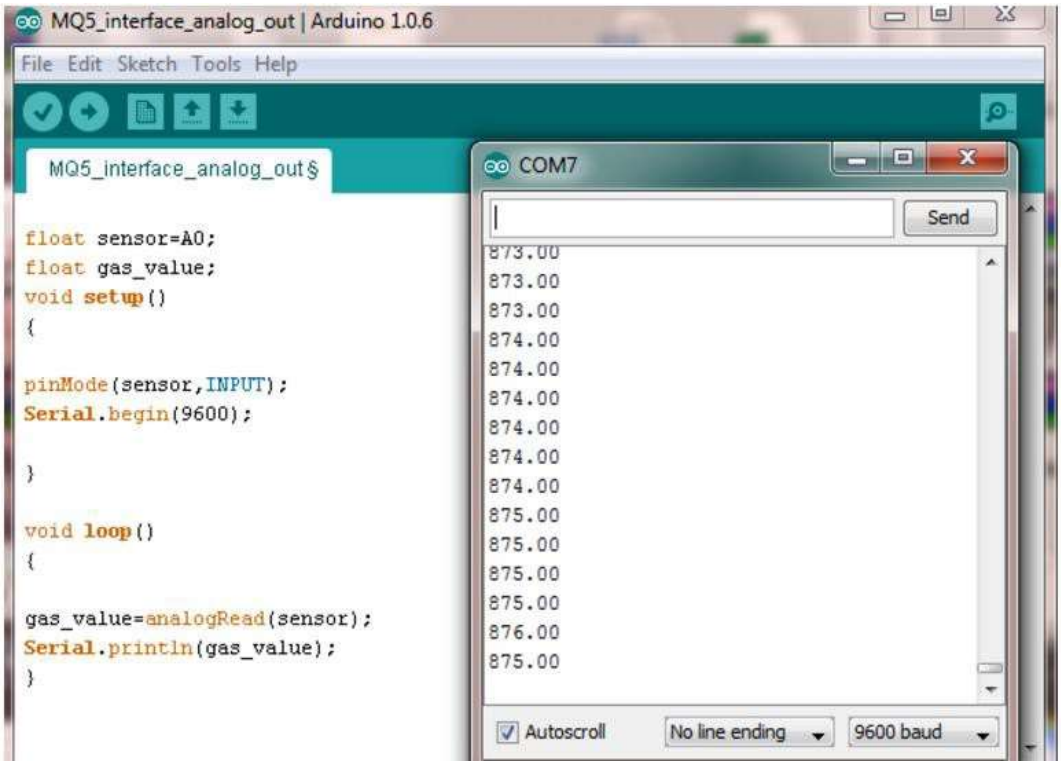
The Below Screenshot Shows Testing Before Applying Gas Leak:

Before applying the gas leakage, MQ135 captures atmospheric air concentration only.



Below Screenshot Shows After Applying Gas Leakage:

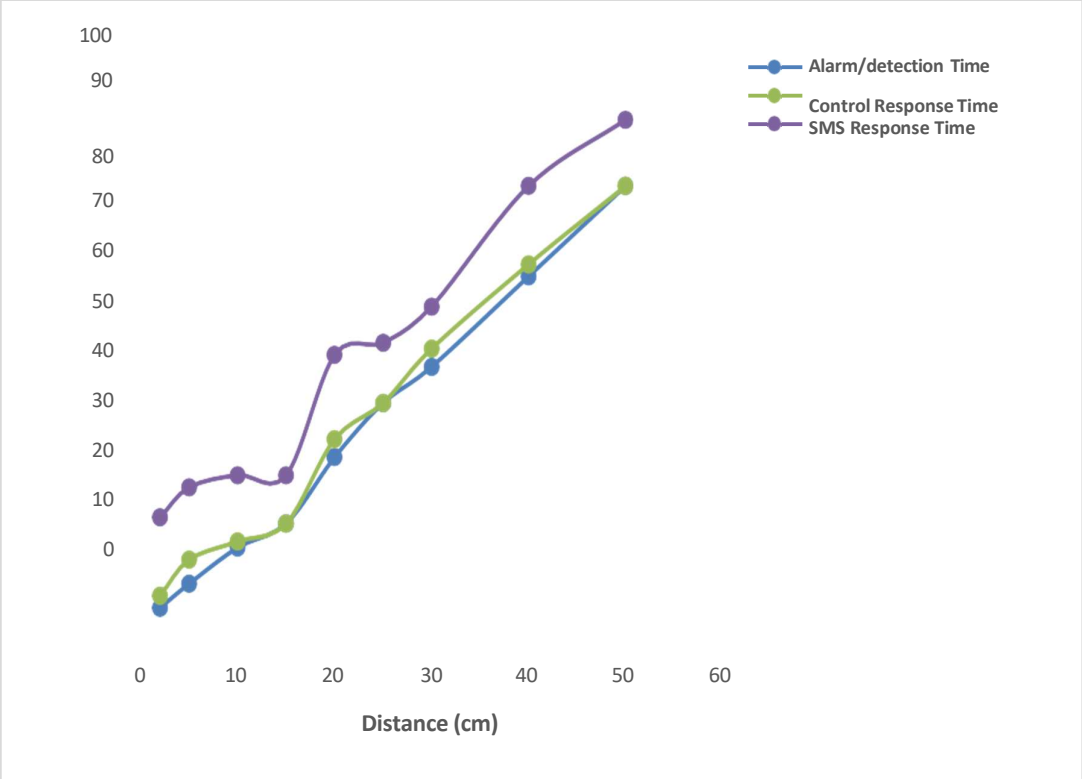
After applying the gas leak, the heating element in the gas sensor will starts heating and varies its concentration value from 1 to its concentration level by pressing the cigar's lightly.



This project deals with the detection of LPG leakage and alert by giving buzzer sound, switching on the exhaust fan and sending an alert message to the person for the precaution to be taken further. For this purpose, gas leakage is sensed by gas sensor MQ2 and sends the information to Arduino by that firstly it switches ON the fan and buzzer to give the alertness in the industry or home. If it continues it again check the threshold gas value and sends the message to the user. To prevent the gas leakage accidents.

In other to obtain a guided decision, two prototypes of the gas leakage and temperature monitoring system were developed. These systems were developed to observe the performance of each setup based on the choice of component used and suggest what system is best for the environment the experiment was carried out. One of the versions of the system [25] uses MQ-6 gas sensor while the newer system utilizes MQ-5. In many similar projects for gas leakage detection, the MQ-series sensors have been tried separately with different microcontroller setup like in [6] they explained the strength and weakness of the MQ-2 sensor when stretch to limit. Meanwhile in [26] MQ-9 was used to test LPG leakage, drawing attention to detection time and concentration of the gas, the project uses only alarm as the means of alert. In our work we incorporate three monitoring and one control devices while utilizing two different sensor types, we aim to focus on time detection with respect to distance and finally to the delay in response of the control system as shown in the graphs (Figure 6) below. The evaluation begins in a chemical Laboratory where substantial amount of heating is performed using Propane Gas Cylinder, this serves as our testing environment. The systems were tested by bringing a propane gas cylinder close to the sensors and continue to vary the distance while data are being collected to observe the detection time and distance. The time and distance were measured using stop watch and meter rule respectively. The data for the time of response of the monitoring unit were also observed using different stop Watch and recorded as shown in the graph below. The sensor detects the LPG gas and sends a signal to the microcontroller, it then analyses the received signal. After that, the microcontroller sends an active signal to other externally connected devices, as a result, a buzzer rings which indicate the presence of the gas. Simultaneously the DC motor opens the ventilation windows and the GSM module send an SMS to a designated mobile phone. Also, when a high temperature is detected using the DHT22, it sends an active signal to the microcontroller and a message is displayed on the LCD screen showing the temperature reading. Upon intervention, when the reset button is pressed, the system refreshes itself and whole system regains a position we refer as the initial state.

The system was tested for 20 minutes as shown on Figure 7 and also at different days. During the test, in order to obtain near accurate timing, the team use different stop watch to monitor the Alarm/detection time, the Control response time and the SMS alert time over the varying distance use. Although the only entity that depends on the distance is the Alarm/detection time, the remaining actively rely on the detection signal as indicated in the graph (Figure 6). A Significant delay was seen for the SMS alert time while difficulties were experienced between recording the detection time and control response time, as in many cases the relay is triggered within seconds of detection time. This performance was also noticed with the second system setup as the only difference was the sensor but in future work, in other to get a better judgement, a complete setup change will be considered. Due to similarities recorded in the reading between the two systems the result of a single system was resolved to be displaced as shown in the graph (Figure 6). In the graph it can be observe that the instability arises mostly from the SMS alert time but overall, the time interval is acceptably fast.



The graph of Gas Alarm/Detection Response

7. CONCLUSIONS

CONCLUSION

When designing a combustible gas safety monitoring system for oil and gas petrochemical or other application, a thorough analysis of application's unique field environment is needed to ensure optimal performance, safety, reliability and cost-effectiveness. A quick decision, of course, can lead to poor detector choices as well as safety, performance, maintenance, and life-cycle cost consequences. As student is doing continuous research on infrared technologies and its application, she believes that gas detection method using infrared is one of reliable method to be used as a precaution to danger that can be caused by gas leakage.

The implementation of automated safety systems such as shutdown valve and alarm system will enhance the effectiveness of this method to users. The integrated infrared circuit has been constructed and tested until this period of time. Future works that has to be done in this project is to fabricate the gas chamber as well as combining the alarm circuit into the integrated infrared circuit. The analysis to measure the reliability of the infrared circuit also has to be improvised so that this gas detector becomes immune to any eternal noise and errors.

In Conclusion, although there are many gas detectors that are better than our project in term of technology but they are frequently used in factory or oil and gas site. Compared to our project, it is specific to use at home. From previous research, there is no safety system that is install at home. Since there are a lot of cases regarding hundreds of houses caught on fire throughout the years, so the early stage before the fire occur must be avoided. This project is produced to detect any leakage of cooking gas and it will alert the user about leakage. User will get the alert in SMS. By using this project, it will reduce the accident of fire and explosion. It also helps the early detection of gas before the concentration of gas reaches the dangerous level.

When designing a combustible gas safety monitoring system for oil and gas petrochemical or other application, a thorough analysis of application's unique field environment is needed to ensure optimal performance, safety, reliability and cost effectiveness. A quick decision, of course, can lead to poor detector choices as well as safety, performance, maintenance, and life- cycle cost consequences. As student is doing continuous research on infrared technologies and its application, she believes that gas detection method using infrared is one of reliable method to be used as a precaution to danger that can be caused by gas leakage. The implementation of automated safety systems such as shutdown valve and alarm system will enhance the effectiveness of this method to users.

Future Scope of the Project

Major cities of India are pushing Smart Home application, gas monitoring system is a part of Smart Home application. Enhancing Industrial Safety using IoT. IoT turns drone into gas detection sensor. Another major future scope could be including an Automatic Shut-off device which will turn off the gas supply whenever it will detect any gas leakage. This system can be implemented in Industries, Hotels and wherever the LPG cylinders are used. This system can be used in industries involving applications such as Furnace, Boilers, Gas welding, Gas cutting, Steel Plants, Metallurgical industries, Food processing Industries, Glass Industries, Plastic industries, Pharmaceuticals, Aerosol manufacturing. As hospitals require to provide maximum possible safety to patients, this system can be used to keep track of all the cylinders used it in Some of the cylinders used are Oxygen cylinder, Carbon dioxide cylinder, Nitrous oxide cylinder. As many students are naïve the risk of causing accidents is high. Hence, our system can also be used in schools, colleges. Many colleges have well established labs including chemistry lab and pharmaceutical labs where gas burners are used. Plenty of medical equipment requires gas cylinders

Thus, a smart helmet for hazardous event detection, monitoring the surrounding environmental conditions and updating information like GPS location and sensor data to the central console for easy tracking and providing oxygen supplements to avoid the inhalation of poisonous gases is designed. The system can also be further developed with the implementation of Internet of Things (IOT). The database can be created that monitors the sensor modules continuously.

REFERENCES

[1]. In the year 2011, A. MAHALINGAM, R. T. NAAYAGI,1, N. E. MASTORAKIS, “Design and Implementation of an Economic Gas Leakage Detector”.

[2]. For Arduino Programming: forum.arduino.cc

[3]. “Design and Implementation of an Economic Gas Leakage Detector” A. MAHALINGAM, R. T. NAAYAGI,1, N. E. MASTORAKIS§ Department of Engineering Systemsschool of Engineering, University of Greenwich (Medway Campus) Chatham Maritime, Kent ME4 4TBUNITED KINGDOM, article in Recent Researches in Applications of Electrical and Computer Engineering

[4]. Johansson, A.; Birk, W.; Medvedev, A., “Model-based gas leakage detection and isolation in a pressurized system via Laguerre spectrum analysis”, Proc. of IEEE International Conference on Control Applications, pp. 212-216, 1998.

Bibliography:

- https://www.google.co.in/
- https://www.youtube.com
- https://how2electronics.com/
- https://www.academia.edu/
- https://en.wikipedia.org/
- https://www.geeksforgeeks.org/
- https://stackoverflow.com/

11. USER MANUAL

Gas Leakage Detection System User Manual

This concise user manual guides end-users with minimal technical knowledge on operating the Arduino-based Gas Leakage Detection System, designed to detect combustible gases (e.g., LPG, propane) and alert users via buzzer, LED, SMS, and exhaust fan activation.

1. System Overview

The system uses an MQ135 sensor to detect gases, providing:

- Audible (buzzer) and visual (LED) alerts.
- SMS notifications via GSM module.
- Automatic exhaust fan activation. It's ideal for household use near gas cylinders.

2. Setup Instructions

1. **Unbox and Verify**: Ensure you have the assembled system (Arduino UNO, MQ135 sensor, GSM module, buzzer, LED, exhaust fan, relay, 5V power adapter) and a SIM card with SMS balance.
2. **Install SIM Card**: Insert a valid SIM card into the GSM module (SIM800L).
3. **Power On**: Connect the 5V adapter to the Arduino. If needed, provide 3.7–4.2V to the GSM module. Place the system 0.5–1m from the gas source, 30–50 cm above the floor.
4. **Calibrate**: Allow the MQ135 sensor to preheat for 1–2 minutes in clean air.

3. Operating Instructions

- **Monitoring**: The system continuously checks for gas leaks.
- **Alerts**: On gas detection, the buzzer sounds, LED lights up, fan activates, and an SMS ("ALERT!! GAS LEAKAGE") is sent.
- **Response**: Ventilate the area, turn off the gas supply if safe, and evacuate if needed.
- **Maintenance**: Check SIM balance, clean the sensor, and recalibrate every 3–6 months.

4. Data Validations

- **Gas Detection (MQ135)**: Requires three consecutive HIGH readings (300 ms) to confirm gas presence, preventing false positives. Calibrated in clean air; recalibrate every 3–6

months.

- **GSM Module**: Checks network via AT commands, sends SMS only once per event to save balance.
- **Power**: Assumes stable 5V (Arduino) and 3.7–4.2V (GSM). System failure indicates power issues.

5. Troubleshooting

- **No Power**: Check 5V adapter connection.
- **No SMS**: Verify SIM balance and GSM power.
- **False Detection**: Recalibrate sensor in clean air.
- **Fan Failure**: Check relay (pin 7) and fan power.

6. Safety Precautions

- Place away from heat/water.
- Test with minimal gas in a controlled setting.
- Use recommended power supplies.
- Prioritize evacuation over troubleshooting during a leak.